



UNIVERSITY OF
SASKATCHEWAN

College of Engineering

Department of Mechanical Engineering
University of Saskatchewan
57 Campus Drive
Saskatoon, Sk
Canada S7N 5A9

Confidential to IMII

RD2019-02 Automated Scaling Bar Phase 2 Final Report

2022-02-18

Travis Wiens, Ph.D., P.Eng
Associate Professor

Josh Rohs
Research Assistant

Department of Mechanical Engineering
University of Saskatchewan

t.wiens@usask.ca

Executive Summary

This project is the second phase of a project with an aim of automating the process of interpreting scaling bar roof impacts, for the purpose of determining mine roof safety. This report presents the electronic, hardware and software design for a compact scaling-bar-mounted device capable of recording a scaling bar impact, transmitting the recording to a phone or table for analysis, classifying the data as “drummy” or “tight”, and displaying the results to the user.

We recorded 9923 impacts over the course of seven days at Nutrien’s Lanigan mine and Mosaic’s K3 mine. The classifier algorithm showed similar accuracy between data recorded on the new device and Phase 1 data recorded using cumbersome studio audio recording equipment. The algorithm also showed similar accuracy between the new data recorded at the Lanigan and K3 mines.

The 3D printed plastic casing and mount design showed good durability, with the exception of the flexible suspension intended to isolate the device from the bar’s mechanical vibrations. The durability of this suspension improved through a number of iterations during the course of this project, but is probably not yet sufficiently robust to long-term use.

Overall, this phase of the project has met its objectives.



Introduction and Background

In Saskatchewan underground potash mines, the condition of the rock in the overhead roof is of critical importance, as rock falls can cause injury or death of workers and significant equipment damage. The roof of a potash mine is generally designed to be composed of a thick layer of consolidated salt (KCl and NaCl) which is structurally strong. Weaker clay layers can intrude on this strong salt beam, weakening it and introducing the possibility of large or small falls of material. The most common method of evaluating the stability of the roof is to strike the roof with a scaling bar (a steel or aluminum bar, usually 2-3 m in length) and to listen to the resulting impact. If the roof beam is solid and firmly attached to the overlying material the impact will sound solid and is termed “tight”. If there are loose layers in the roof, they will vibrate separately from the overlying mass of material, causing a reverberating sound which is termed “drummy”. If drummy ground is detected, a roof bolting procedure is triggered to attach it to the overlying material.

One issue with this method is that it is subjective: novice workers may not be able to confidently differentiate between tight and drummy impact sounds and may either fail to trigger ground support procedures were warranted (a safety hazard) or may erroneously call for unnecessary ground support (causing production delays). The overall goal of this project is to automate the collection and evaluation of the roof condition, reducing the subjective nature of the procedure. An additional advantage of this project is that the system may be used autonomously; for example, automatically impacting and evaluating the roof as the mining machine progresses forward.

In a previous study [1], the feasibility of this method was established. A large number of impact recordings were taken from a variety of mine sites using a tall microphone stand, digital audio recorder and studio microphones. This data was analyzed offline using machine learning algorithms and it was determined that there was enough information in the recorded audio to reliably classify the recordings as drummy or tight. However, the implementation was not practical as it required two people to perform the impact and record data, and the results were not immediately available.

The goals of this phase of the project were to produce a practical device to implement this method. The device was envisioned as a small box that mounts on the scaling bar, containing microphones and data acquisition hardware as well as computation power to analyze the recorded data and communicate the results to the user. This would also have Bluetooth communication with a worker’s tablet or phone for configuration and display of details.



Equipment Design

Circuit

The automatic scaling bar (ASB) firmware runs on a ST Micro STM32WB module. This device includes a microcontroller as well as embedded bluetooth capabilities, making it simple to control from a mobile phone or tablet. The processor mainly handles user interface and audio acquisition/processing. A real time clock (RTC) is included to allow for timestamping of recordings. Although the STM32WB module was selected, the circuit board has space allocated to allow for other modules to replace it (e.g. ESP32)

The main peripheral devices are the pair of microphones, Vesper VM3011. These are MEMS devices with built-in amplifiers and data acquisition modules, simplifying the circuit design and reducing the opportunity for noise interference. The microphones are mounted on either end of the case. This allows for the case to be mounted in either direction and the second microphone (pointed away from the impact) can be used to reduce the effect of room acoustics and structural vibrations. Data was recorded at a 48 kHz sample rate and written to an SD card.

Although the primary method of control and display of result is expected to be via a tablet or smart phone, the circuit includes two RGB LEDs and two push buttons that would allow for basic operation when not connected.

Although not yet implemented, the design also includes an accelerometer and speaker. The accelerometer can be used for vibration compensation or to trigger recording. The speaker may be used to give an audio cue corresponding to the classifier's result.

The device is powered by a 2200 mAh Lithium ion battery that is charged via the USB port. Although not experimentally verified, this battery is expected to allow for up to month of operation between charges. The circuit design enhances battery life by powering each peripheral device via the microcontroller, so they may be powered off when not in use. The battery's state is monitored by the microcontroller and can be reported to the user via Bluetooth.

Photos of the final circuit board design are shown in Figure 1 and 2.



UNIVERSITY OF
SASKATCHEWAN

College of Engineering

Department of Mechanical Engineering
University of Saskatchewan
57 Campus Drive
Saskatoon, Sk
Canada S7N 5A9

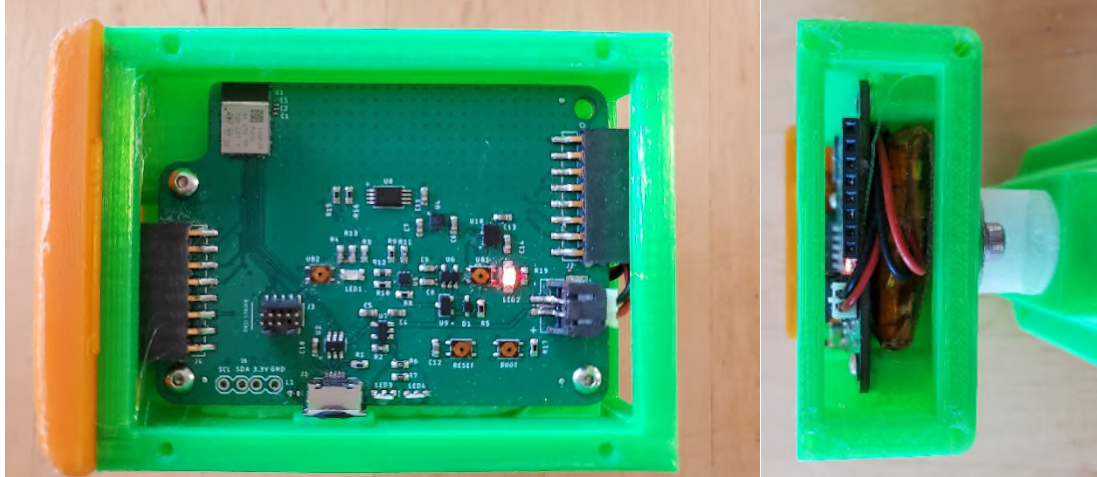


Figure 1: Main circuit board top view (left) and side view (right). The battery is visible below the board in the side view.



Figure 2: Microphone daughter board. The pins on this board mate with the headers on the main board.

Casing

The casing, shown in Figures 3 and 4, is intended to provide IP66 protection from dust and liquids, as well as being robust enough to survive the mine environment which may include rock falls. We selected 3D printing to produce the case, as this would allow for an optimized custom design and fast design iterations. This production method was selected for the prototype but may also be appropriate for a production run, considering the small quantities expected. The casing was produced using PETG material



on a Fused Deposition Modelling (FDM) printer. The plastic casing allows for Bluetooth signals to pass through without the need for an external antenna. Dust and liquid ingress is prevented by IP68 USB port and plastic film over the microphone port and buttons. The device mounts to the scaling bar using a two clamps, which can be customized to fit the bar's cross section without modification to the rest of the casing.

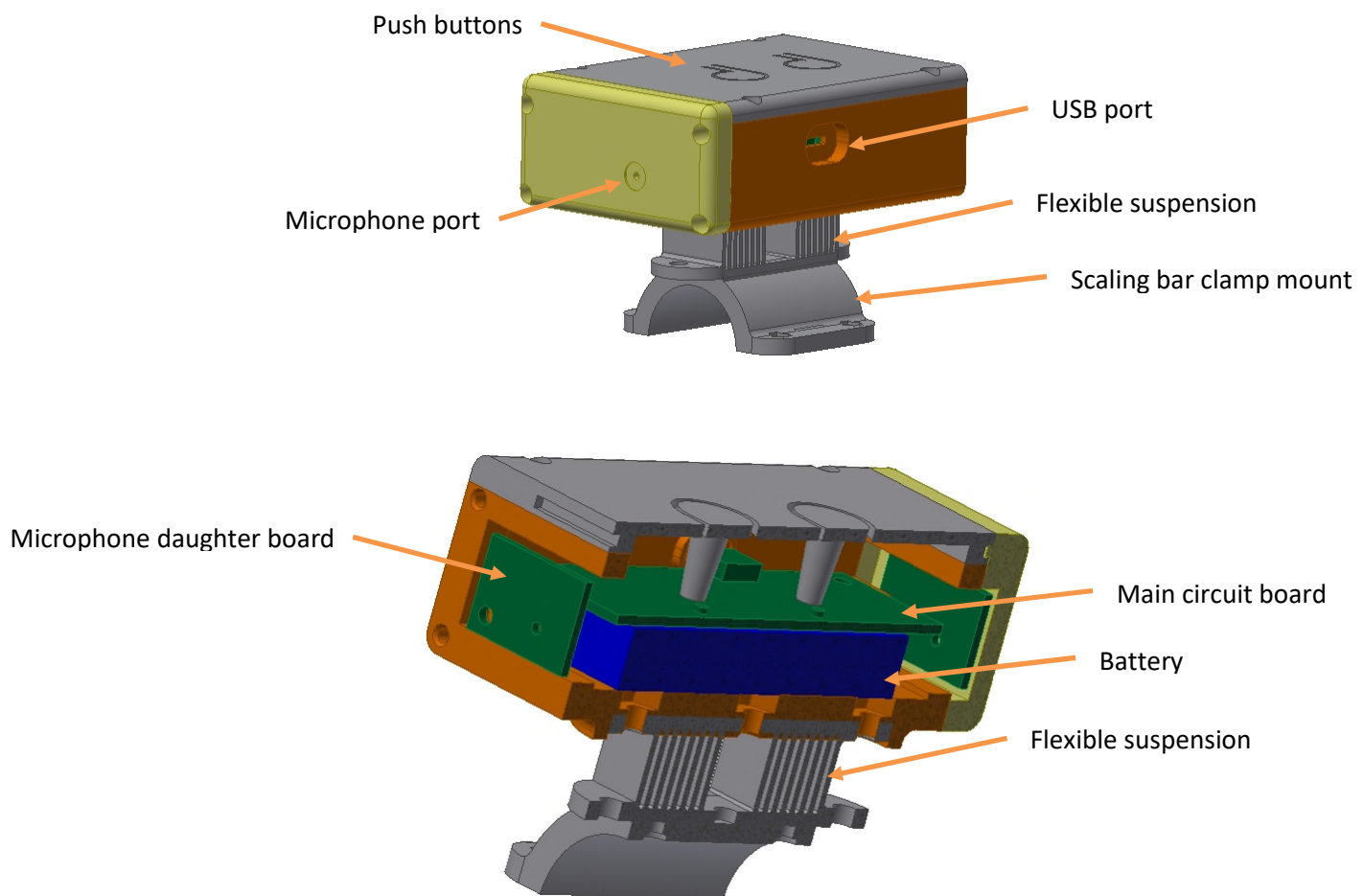


Figure 3: Casing design and cutaway.



UNIVERSITY OF
SASKATCHEWAN

College of Engineering

Department of Mechanical Engineering
University of Saskatchewan
57 Campus Drive
Saskatoon, Sk
Canada S7N 5A9

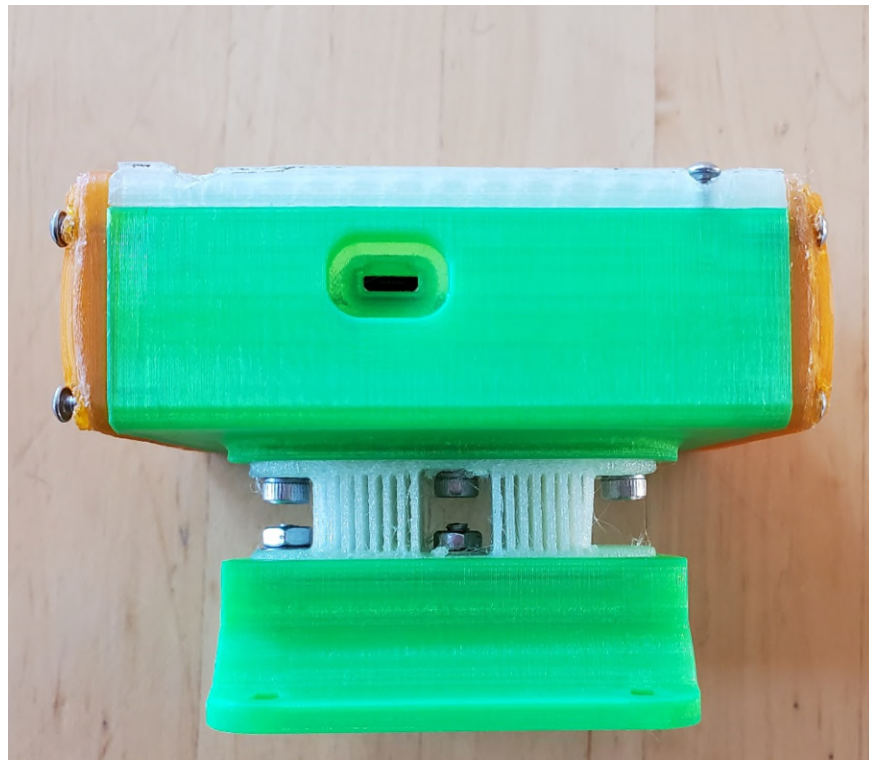
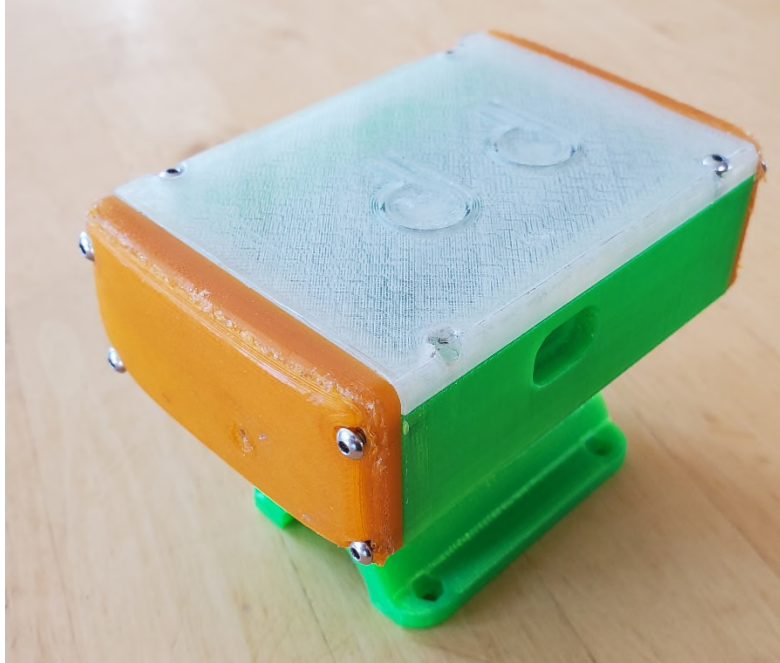


Figure 4: Photos of the assembled device.



Software Design

There are two pieces of software involved in this design: the firmware that runs on the device's microcontroller and an app that runs on the user's phone or tablet.

The microcontroller firmware is responsible for

- Detecting an impact
- Recording the impact audio
- Allowing the user to label an impact as “drummy” or “tight” (optional)
- Saving the timestamped and labelled recording to SD card (optional)
- Transmitting the recording via Bluetooth to the app
- Receiving classifier results from the app
- Displaying the classifier results via the RGB LED
- Monitoring battery life and communicating this to the app

Although not fully implemented, the software allows for the classification algorithm to optionally run on the microcontroller, eliminating the need for a Bluetooth connection to the app. However, the classification algorithm will run slower and use more battery life.

The app (running on the phone or tablet) is responsible for

- Sending configuration options to the firmware
- Triggering impact recordings
- Displaying classification results
- Allowing user to label recordings
- Displaying battery life

This was implemented on an Apple iPhone or table. Some screenshots are shown in Figure 5



UNIVERSITY OF SASKATCHEWAN

College of Engineering

Department of Mechanical Engineering
University of Saskatchewan
57 Campus Drive
Saskatoon, Sk
Canada S7N 5A9

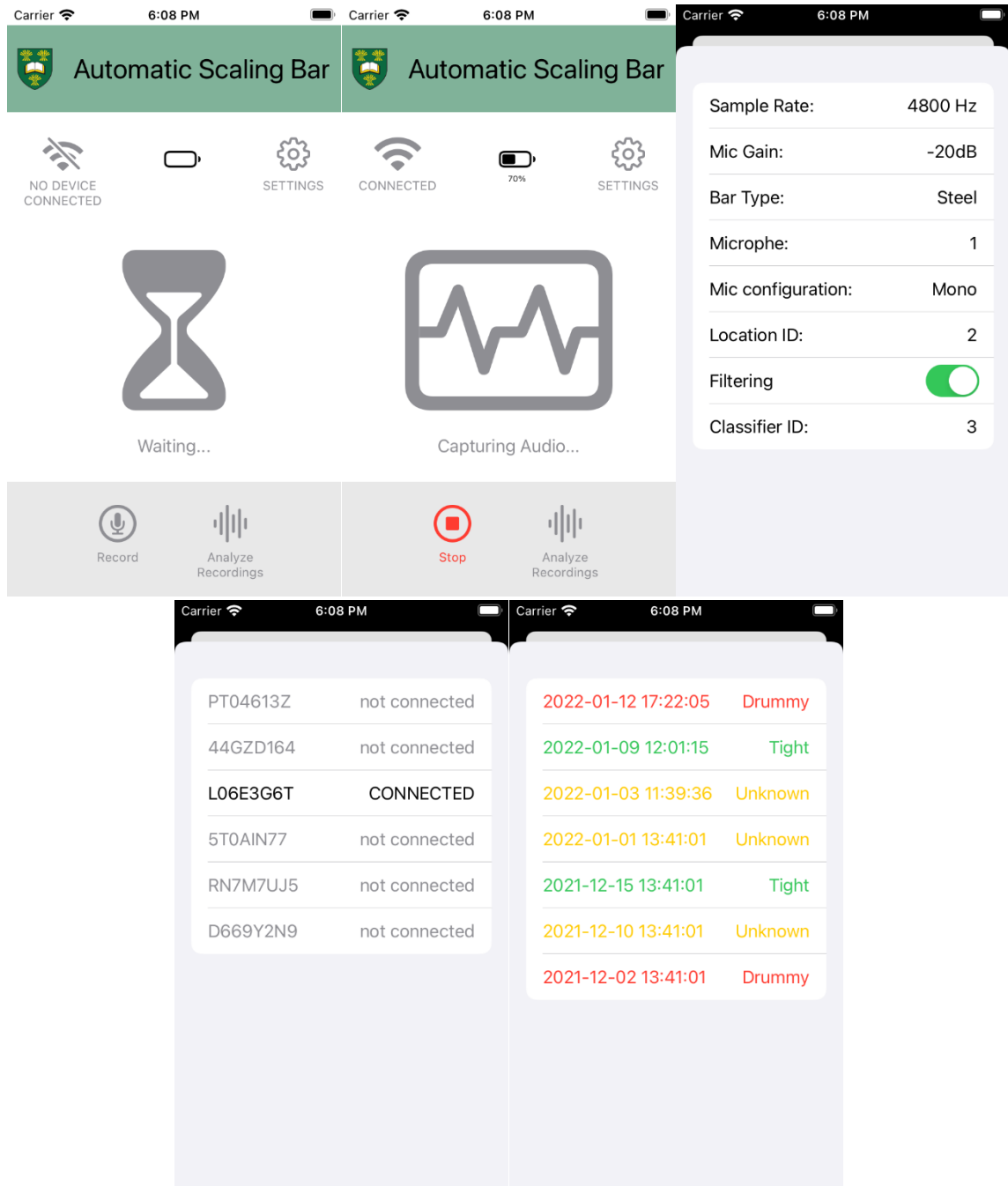


Figure 5: Screenshots of sample app screens.



Classifier Algorithm

The classifier algorithm is responsible for determining whether a selected recording should be reported to the user as “drummy” or “tight”. We used an ensemble method to perform this classification. This is when a large number of “weak” learning algorithms are pooled to produce a “strong” learner. As long as each weak learner performs the classification more accurately than chance, then the aggregate can be shown to converge to a high level of accuracy. In our case, we used an ensemble of 400 decision trees, optimized using Matlab’s GentleBoost algorithm [2]. This is recommended as an algorithm that is resistant to overfitting, which is a concern in our case due to a limited number of data samples.

Each recording was preprocessed by normalizing the signal power to remove differences due to variation in how hard the operator struck the roof. The start of the signal was identified as the first point that was over 10% of the standard deviation of the full 0.67 s recording. We then selected 100 samples prior to this point and 4700 following it (0.1 s total). This time may have to be adjusted in particularly reverberant rooms, but it is believed that most of the useful information comes in the initial part of the recording. This time domain model was concatenated with the magnitude of its Fast Fourier transform to form the input to the classifier. The expert’s labelling of “drummy” or “tight” was encoded as 0 or 1 respectively for the binary classifier.

Testing Results

Recording Acquisition

A total of 9923 impacts were recorded over seven days at two mine sites: Nutrien’s Lanigan Mine and Mosaic’s K3 mine. Recording locations were selected to cover a representative sample of roof conditions in each mine. Each recording was labelled as “drummy” or “tight” by a expert mine employee. Scaling bars of varying cross section and material (steel and aluminum) were tested.

Since it is unsafe to enter areas of unsupported drummy ground, the drummy recordings were taken from areas that had been supported by roof bolts. It is assumed that the roof bolts do not significantly affect the impact sound.

Ease of use

In Phase 1 of this project, as digital audio recorder, tall microphone stand and studio microphones were used to record impacts. This procedure was slow to collect data, with an average of 552 recordings acquired per working day [1]. In Phase 2, the new device is much more conducive to recording bulk training data and the acquisition rate was increased to 1418 recordings per day (9923 over seven days), largely limited by travel time. It is believed that sufficient recordings to characterize a new mine location can be achieved in 1-2 days.



Classifier Accuracy

We evaluated the accuracy of the classifier using a K-fold validation method. This means that the dataset is divided into a number of sections (10 in our case). The classifier is then trained using nine of these sections and then its accuracy is evaluated on the remaining section (using data not seen in the training). This is then repeated with each section sequentially being used for validation. The ten classifier error values can then be used to obtain a mean classifier error and also can be used to calculate the uncertainty in this mean.

The two main goals of this analysis were to determine whether the hardware device and its MEMS microphone could match the accuracy of the data recorded using studio audio gear in Phase 1, and to determine whether the method would work at the K3 mine, as the Esterhazy Member has a different geology from the Patience Lake Member that was studied in Phase 1. Figure 6 presents the Phase 1 data for the Lanigan mine, which can be directly compared to the Phase 2 data from the Lanigan mine. It also shows the new data for the K3 mine. This data shows that the Phase 1 and Phase 2 Lanigan data are statistically indistinguishable, as are the Lanigan and K3 Phase 2 data. This verifies the acoustic design of the new device and provides evidence that the concept is feasible in the Esterhazy Member.

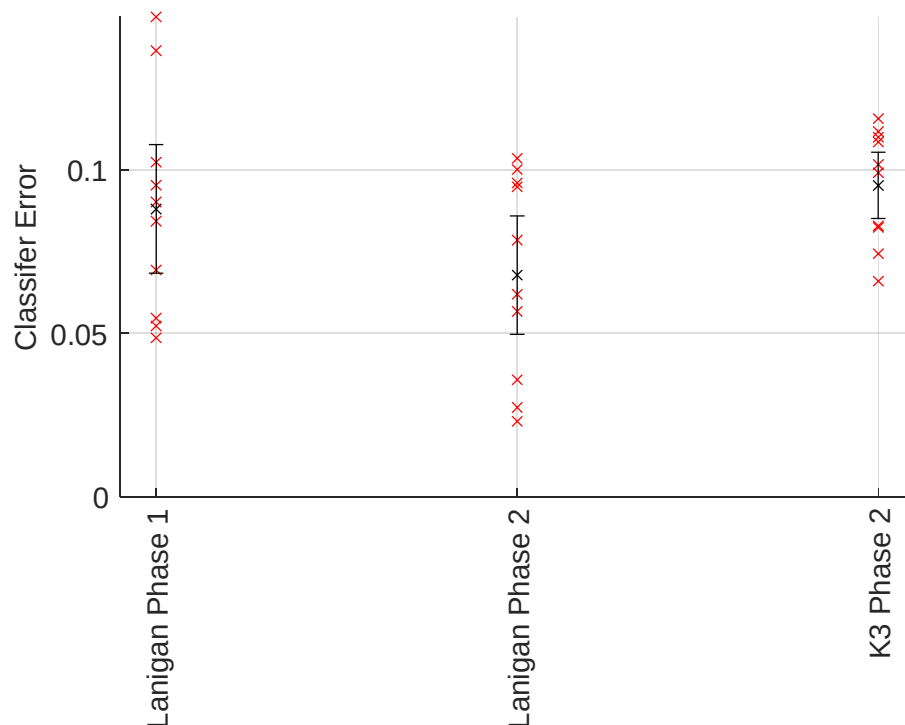


Figure 6: Classification error for Phase 1 (using studio audio equipment) and Phase 2 using the new compact device, for Lanigan and K3 mines.



Signal to Noise Ratio

We quantified the signal-to-noise ratio (SNR) for this dataset by measuring the signal power in the impact signal (as selected using the algorithm in the previous section) and the signal power in the part of the recording prior to the impact. A histogram of this data is shown in Figure 7. The Lanigan data had a median SNR of 54.9 dB and the K3 data had a median of 61.0 dB. These are relatively high SNR values and demonstrate that there is not excessive self-noise in the microphone. Most of these recordings were taken in a quiet area of the mine with no equipment running, so represent a best-case scenario, but this should be a common situation as roof soundings are typically performed before starting equipment. However, this may be an issue if the system is used to automatically evaluate the roof above a running miner, as the accuracy will be degraded as the SNR is reduced.

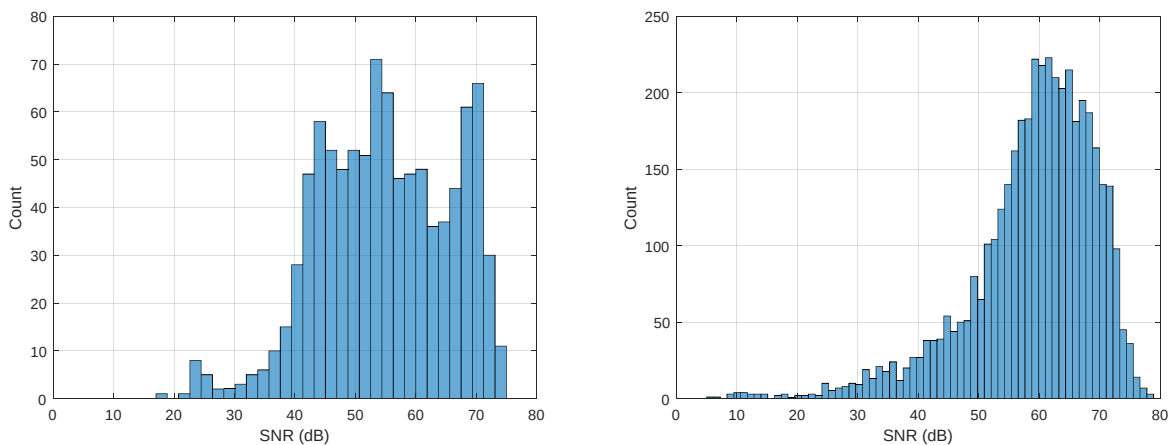


Figure 7: measured signal to noise ratio for Phase 2 recordings at Lanigan (left) and K3 (right).

Battery Life

Although the battery state of charge was not measured, the device was used to record up to three days of continuous data acquisition between charges without a low battery condition. As normal use would represent a much lower usage rate (testing a few impacts when entering a room or at start of shift and then waiting for hours), it is not unreasonable to expect a single charge to last weeks.

Suspension and Casing

One objective of this testing was to evaluate the durability and robustness of the hardware. Although the seven days spent testing the equipment only represents a fraction of its expected life, it can demonstrate obvious faults. The 3D printed enclosure showed no signs of damage during this time and seemed robust. There was no damage to the enclosure's membranes sealing the microphone port or switches and there was no indication of plugging by dust.



The flexible suspension that was used to isolate the enclosure from the bar mount did show wear in initial tests. Early iterations suffered from tear-out of mounting screws. Later iterations with larger and more numerous screws as well as a longer suspension improved this to the point that the device was useable for a number of days, but may not be suitable for the entire expected life of a unit. This is an area for future improvement.

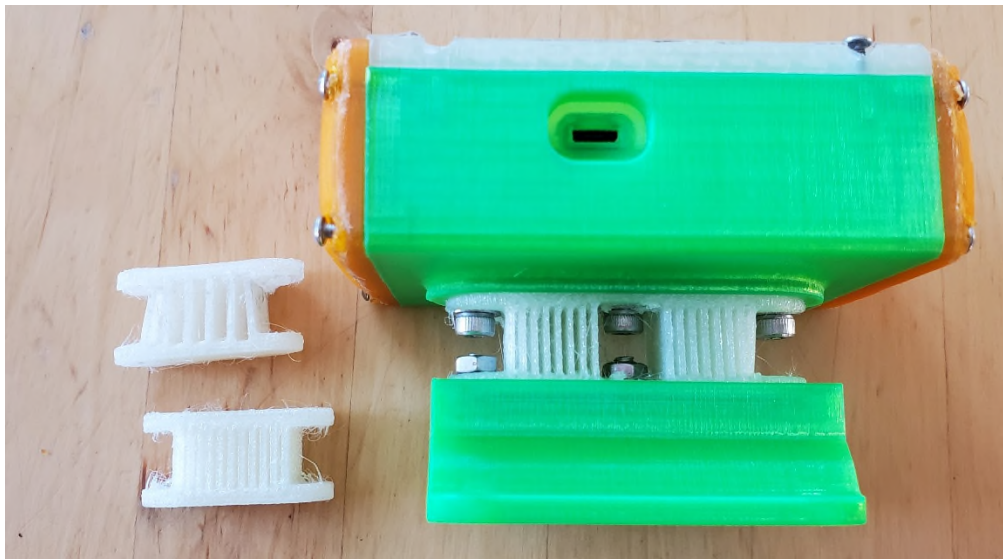


Figure 8: two previous, failed suspension designs (left), along with the present design (right).

Steel vs Aluminum Bars

We noticed that the scaling bar material had a significant impact on the recordings. The steel bar had more structural reverberations, with the decay time and frequency overlapping with the acoustic impact signal for drummy ground. This prevented the use of heuristic classification methods (such as using decay time) and can be expected to also degrade the performance of the machine learning classifier (although we did not quantify this effect). It was also noticed that the stronger vibrations in the steel bar had a tendency to loosen the mounting hardware, causing an interfering rattling sound. The aluminum bar was less stiff, so this effect was considerably less pronounced. This effect could also be mitigated by reducing the coupling of structural vibrations to the microphone via improved suspension design or by using the second microphone to cancel some structural vibrations.



Recommendations

This phase of the project has demonstrated the feasibility of a compact, scaling bar-mounted device.

Next steps in the development of this device include:

- Continuous improvement of the classification algorithm
- Improvement to the flexible suspension, considering both durability and vibration transmission reduction
- Verification of the long-term durability of the casing
- Testing of the app useability

Longer term development towards an automated solution would include

- Testing of the algorithm in a noisy environment (e.g. mounted on a miner)
- Development of an automated striking device

Conclusions

In this phase of the project, we have:

- Developed, produced, and tested an electronic design capable of acquiring, storing and transmitting impact recordings
- Developed, produced and tested a mechanical design including case, suspension and scaling bar mount.
- Continued development of the classifier algorithm
- Verified that the acoustic design matches the performance of the studio audio equipment used in Phase 1
- Verified that the concept works in the Esterhazy member with comparable performance to the previous data recorded in the Patience Lake member

We believe this has been a successful project and has significantly advanced this technology toward commercialization.

References

[1] Md Shahriar Islam, Travis Wiens, 2019, "Feasibility Study of Automatic Classification of Scaling Bar Impacts for Underground Mine Safety" Report to IMII

[2] Mathworks, 2021, "Ensemble Algorithms" <https://www.mathworks.com/help/stats/ensemble-algorithms.html> retrieved 2022-02-18